

# Toward efficient energy systems based on natural gas consumption prediction with LSTM Recurrent Neural Networks

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## ABSTRACT

Finding suitable forecasting methods for an effective management of energy resources is of paramount importance for improving the efficiency in energy consumption and decreasing its impact on the environment. Natural gas is one of the main sources of electrical energy in Algeria and worldwide. To address this demand, this paper introduces a novel hybrid forecasting approach that resolves the two-stage method's deficiency, by designing a Multi Layered Perceptron (MLP) neural network as a nonlinear forecasting monitor. This model estimates the next day gas consumption profile and selects one of several local models to perform the forecast. The study focuses firstly on an analysis and clustering of natural gas daily consumption profiles, and secondly on building a comprehensive Long Short Term Memory (LSTM) recurrent models according to load behavior. The results are compared with four benchmark approaches: the MLP neural network approach, LSTM, seasonal time series with exogenous variables models and multiple linear regression models. Compared with these alternative approaches and their high dependence on historical loads, the proposed approach presents a new efficient functionality. It estimates the next day consumption profile, which leads to a significant improvement of the forecasting accuracy, especially for days with exceptional customers consumption behavior change.

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## 1. Introduction

Natural gas consumption forecasting has proven to be one of the most delicate tasks that power system operators face during the last decades. Energy load prediction in Algeria is highly challenging, due to its substantial European contribution in energy supply, on one hand, and the country size, population and the important economic growth on the other hand. Indeed, the country spans on an enormous area of 2,381,741 km<sup>2</sup> in the north of Africa, between the 19–37° north latitude and 9° west and 12° east longitudes. This has a significant consequence on its climatic and customers behavior diversity, making energy load forecasting even more complicated.

Since 1964, Algeria is amongst the biggest natural gas producers and suppliers in the world, and the energetic flux has never been interrupted. Moreover, the national society of electricity and gas in Algeria (SONELGAZ) has to deal with multiple and important

market restrictions in several domains. In particular, the reserve, storage and production which is in an increasing rate constitute a challenge in the last years for different reasons such as irregular supply and periods with peak consumption (see Section 3.1.1).

The search of optimality in forecasting is strongly driven by economic reasons, as natural gas resources are scarce. The increased demand and competition require new “eco-friendly” technologies. Since the Algerian economy is relying substantially on hydrocarbons exports with a strong economic growth, the availability of optimal forecasting natural gas consumption models, is therefore of paramount importance.

Energy consumption studies differ according to the forecasting horizon and may be categorized into: *long-term forecasting* (3–20 years), *medium-term forecasting* (a month to 2 years) and *short-term forecasting* (30 min to a week) where many different methods have been employed to forecast energy consumption on three basis. In the field of natural gas load forecasting, Soldo [1], provides a detailed overview of related works and a classification based on prediction horizons, model paradigms, geographic areas and other criteria. The last two decades witnessed increased investigations aimed at improving the reliability and accuracy an forecasting

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methods [2]. The research in this area has been conducted by developing three main classes of techniques: statistical time series based techniques (Seasonal Auto-regressive Integrated Moving Average with exogenous inputs (SARIMAX) [3,4], functional autoregressive with exogenous variables [5], vector autoregression [6] and extended Kalman filters [7]), non-linear regression methods (Support Vector Regression (SVR) [8], multivariable adaptive regression splines [9], regression trees [10], etc) and computational intelligence based models (Artificial Neural Networks (ANNs) [11,12], fuzzy logic [13] and support vector machines [14]).

Numerous types of models, suitable for short-term prediction have been developed using a range of tools, approaches and strategies, from linear to nonlinear paradigms including machine learning and neural networks models. The later have been extensively used in the last decades due to their ability to generalize well from a set of clustered input/outputs. Furthermore, Jolanta Szoplik [15] presents results of gas demand forecast in Szczecin, (Poland) using Multi Layered Perceptron (MLP), a type of ANNs. The modeling process consists of considering both calendar and weather factors to build several MLP models, and the one with higher quality is used to predict the gas consumption on any day of the year and any hour of the day. Feng and Xiaozhong [16] also propose a MLP ANN. In Ref. [16] instead of randomly initializing the network weights and thresholds, genetic algorithms are used for this task and the network is trained with an improved Back Propagation (BP) algorithm by including additional momentum and self-adaptive learning rate. Tonković et al. [17] develop an MLP and Radial Basis Function (RBF) to model a 24 h forecasting for the north-east region of Croatia. Taşpinar et al. [3] propose an MLP and RBF ANNs with time series to perform daily natural gas consumption forecasting in some regions of Turkey based on the ambient air temperature, average cloud cover, relative humidity, wind speed and atmospheric pressure as meteorological data.

Since the first successful applications of ANNs to energy load prediction modeling [18,19], many researchers have explored alternative types of ANNs. Recent works have focused on Recurrent Neural Networks (RNN) and recurrent Long Short Time Memory (LSTM) networks. In the same context, Kong et al. [20] have practically demonstrated that LSTM networks achieve generally better forecasting performance in individual residential households short-term load dataset compared with standard BP ANN and the *k*-nearest neighbor approach.

In Potočník et al. [21] a comparison between static and adaptive forecasting models is presented. Various models are studied, including linear models (Multiple Linear Regression (MLR), linear auto-regressive with exogenous inputs and recurrent autoregressive with exogenous input), neural network models (with one hidden layer feedforward ANN and a RNN) and Support Vector Regression (SVR) models. Calendar effects and forecasting with different granularity are studied by Lusis et al. [10]. MLR, regression trees, SVR and MLP ANNs applied to 30-min forecasting intervals, are shown to improve the accuracy of one day-ahead forecasting strategies.

Considering short-term natural gas load forecasting as time series modeling, another noteworthy comprehensive review is presented by Deb et al. [22]. The study analyzes thoroughly the most widely used techniques to forecast historical peak energy time series data. Furthermore, Deb et al. show that hybrid models some of the most efficient techniques for time series energy forecasting, especially for buildings.

Hybrid forecasting techniques have been developed and have practically proven their efficiency by achieving high accurate forecast results compared with single model approaches [23,24]. They have demonstrated their reliability in solving complex non-linear prediction and control problems. Hernandez et al. [25]

propose four MLP models to estimate the peaks and valleys in the electrical load time series, then feed the obtained results together with other variables as inputs into another MLP models to predict the next day's total load. In a similar manner, Krzysztof and Tomasz [26] presented an approach based on classifying the peaks of the time series load using three models with respect to the number of peak levels. ANN, SVM and Random Forest (RF) models are built and tested to accomplish the peaks classification step. In the forecasting step, another ANN, SVM and FR single model is used to forecast the next 24 h load based on the estimated peak. Following the same context, Ilıc et al. [27] present a hybrid structure based on two MLP models, where the first one predicts integrated load value of the next day, the output is then used as input and fed to the second MLP network in order to obtain the forecasts for each of the 24 h for the forecasting day. Ghadimi et al. [28] are interested in forecasting the electrical load and price and propose a multistage hybrid forecasting approach that employs three different models. ANN, RBFNN and SVM predict the load obtained from feature selection process, then combine the three forecasts to generate a single unified price/load forecast with a modified ordered weighted average.

A different, less used, class of hybrid approaches are the two-stage adaptive architectures also known as the divide and conquer approaches. The key idea of the two-stage approaches is: firstly, divide the time series data into several subsets for the purpose of minimizing the non-stationarity recorded in the data and reducing the input space. Secondly, model and predict the time series load in each subset with a separate model. Finally, a chosen technique is applied to integrate the model results. Zhang et al. [29] propose an approach that decomposes the original electricity load into two components, a linear and a non-linear one. Considering the trend as the linear component, the non-linear trend is extracted by subtracting the trend from original electricity load. Furthermore, both components are predicted separately using ARIMA and wavelet ANN, then model results are summed to obtain the final forecasting loads. Substantiating the concept of the superiority of divide and conquer hybrid approaches over the traditional single model ones, Xiao et al. [30] propose a hybrid method to model yearly energy and oil consumption time series which does not include seasonal patterns. The basic idea is to predict the linear trend with an AR model and focuses on the left residual sequence which is the non-linear subseries, then utilises four ensemble prediction models, namely, BP ANN, SVR, GP and RBF ANN. The final decision is made by a special ANN model to establish selective combination forecasting. Thus, both outputs from AR model and the optimal complexity model are added to obtain the final energy consumption value.

In order to perform efficient planning management, it is necessary to best match the daily and weekly seasonality affected by customers' decisions and weather conditions with historical consumption profiles using adequate and accurate models.

### 1.1. Main contributions

In response to the challenges that energy systems face this work proposes a novel hybrid approach and evaluates its efficiency. This paper advances the current state-of-the-art methods and has the following main contributions: *i*) an adaptive hybrid architecture is proposed for natural gas consumption forecasting in a large geographic area. *ii*) LSTM models is proposed that can predict efficiently the natural gas consumption. *iii*) a MLP model is proposed to estimate the next day consumption profile.

Besides the geographical area that has not been studied before, this work focuses in the first place on carefully identified clusters of a daily consumptions. This step known as the consumption profile

identification step divides the dataset into several subsets using K-Means in order to reduce the non-stationarity of the time series. Consequently, a number of local models matching the number of clusters are developed in a process known as the modeling phase. Multiple LSTM ANN models are constructed and assessed according to the nature of daily natural gas consumption profiles included in each cluster with respect to existing exogenous factors, rising the issue of managing and choosing the appropriate model for each forecast. Finally, a forecasting MLP ANN classification model, is designed and trained. After conducting several experiments, one of several local LSTM developed models is selected. The performance of the developed approach is thoroughly compared with a number of benchmark state-of-the-art approaches.

## 2. Outline of the proposed approach

The developed approach consists of several stages and these are shown on Fig. 1. The first stage considers and identifies the customers the consumption profiles. A clustering method is used for the purpose of reducing both of the non-stationarity of the natural gas consumption time series data and the inputs space. The second stage consists in modeling the gas consumption profile. After dividing the data, multiple models are developed according to the nature of each data subset.

This two-stage forecasting approach possesses high learning and prediction capabilities compared with conventional approaches since it is based on a global model and identified clusters, obtained either from the training or test datasets [31]. Despite the effectiveness on minimizing the range of error, one the limitation of this approach is that, there is no certain functional process could determines which model should perform the prediction of the next day, this is the case when attempting to forecast new days which their consumption profile has not been identified.

Besides realizing day type consumption classification and modeling an efficient natural gas consumption with LSTM models,

this paper introduces a novel monitoring stage that resolves the two-stage approach's deficiency, by developing a combinatory MLP used as a Forecasting Monitor (FM). The role of the FM-MLP model lies in estimating the next-day consumption profile according to the similarity measures calculated during the clustering phase, then choosing the right local LSTM model to perform the forecasting for next day.

## 3. Methodology

### 3.1. Day consumption profiles classification

#### 3.1.1. Available data

According to the Annual report from the Algerian ministry of Energy [32], the production of natural gas in Algeria reached 96.6 Bm<sup>3</sup> for the sole year 2017, representing 55% of all primary energy production types.

The total national energy consumption (including losses) reached 59,6 Mtep in 2017. This is equivalent to (760 BTh 76 Bm<sup>3</sup>) reflecting an increase of 2.1% when compared to 2016 dominated by natural gas 37%, electricity 30% and oil product 27% consumptions percentages. This increase, when compared to 2016, is mainly driven by the increase of the electricity consumption (5.5% increase). More than 90% of electricity production in Algeria is thanks to natural gas driven technologies [33]. Natural gas consumption witnesses a 1.4% increase, and GPL (Liquid natural gas, mainly as fuel vehicle for transport purposes) with a 5% increase.

It is clear from the above statistics that energy consumption is due to social economical growth of the country mainly driven by the domestic demand (44%) with a total number of energy subscribers of 9,2 millions against 8,8 millions at the end of 2016. Liquid gas also has a demand increase, mainly driven by GPL/C (liquid gas for vehicles) with an increase of 30% due to the promotional price of \$0,076 per liter. Table 1 shows the evolution over the last two years of national natural gas consumption in m<sup>3</sup> and

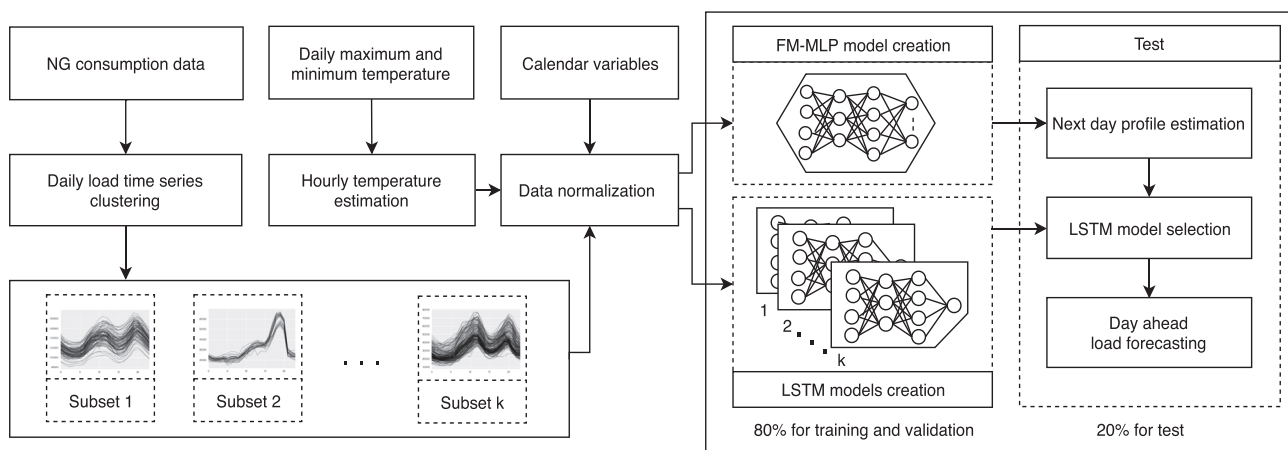


Fig. 1. Main steps of the proposed approach.

Table 1  
Natural gas consumption over the last two years.

| Product                                              | Units             | 2016   | 2017   | Evolution |      |
|------------------------------------------------------|-------------------|--------|--------|-----------|------|
|                                                      |                   |        |        | Quantity  | %    |
| Natural Gas                                          | 10 m <sup>3</sup> | 22 997 | 23 311 | 297       | 1,4  |
| Electricity                                          | GWh               | 70 748 | 75 675 | 932       | 5,5  |
| Estimation of natural gas for Electricity production | 10 m <sup>3</sup> | 17 828 | 19 070 | 1242      | 5,5  |
| Total gas consumption                                | 10 m <sup>3</sup> | 40 825 | 42 381 | 1539      | 3,63 |

electricity in GWh. The amount of natural gas needed to produce electricity power is also calculated, taking into account the average specific consumption factor of steam and combined cycle turbines (approximately equal to 2.8), which constitute the basis of electricity power generators.

If the national demand trend in energy is continues in the next few years, the national production in natural gas will barely suffice to cover it. Consequently, the export capabilities of Algeria by 2030 will be reduced to 10 Bm<sup>3</sup>, which could lead to a critical situation for an economy relying on exports of fossil resources [34].

The present approach is validated over Algerian hourly natural gas consumption data for both residential and industrial sectors for 2014 provided by the national electric and gas company SONELGAZ. However, hourly temperature data are missing, with only minimum and maximum daily temperatures provided. In the lack of real hourly temperature measurements, linear and cubic interpolation fails to represent the daily temperature profile dynamics. For this purpose daily hourly temperature profiles are approximated using the minimum and maximum daily values recorded by the Algerian meteorological office using the equations introduced by Linvill [35]. Daily generated temperature profiles [36] are used as exogenous inputs for the prediction of natural gas load. Fig. 2 presents hourly natural gas consumption for the residential and industrial sector and generated daily temperature profiles. Moreover, the load fluctuation through the entire observed year shows a strong dependence with the time of the year (month) as well as the temperature.

Beside temperatures, natural gas consumption variability over time is dependent on different essential exogenous variables when considering LSTM neural networks modeling. The most influential input variables could be grouped in: historical factors, meteorological factors (such as temperature, wind speed, humidity and sunshine) as well as economic factors (oil prices, number of clients, GDP and natural gas price). Considering short term forecasting, on an hourly basis, it is highly recommended to take into account other types of factors in order to cover the seasonality pattern, usually in the form of one hot encoding inputs (eg., hour of the day, day of the week and month of the year) [37].

Due to the non-stationarity in historical gas consumption time series, the daily load time series are regrouping using a clustering

method. Based on the results obtained from the clustering phase, each day in the dataset is labeled and grouped with the other days that have a similar curve according to the clustering method measures. A local forecasting approach based on multiple models is adopted in Ref. [38], where each model is constructed to focus on a single clustered subset. Usually this approach outperforms global model-based forecasting approaches, where a single model handles the forecasting process for the entire output space.

The first stage of the proposed approach is to classify samples of historical natural gas load segments from the year 2014 with:  $H_{2014} = \{d_0, d_1, \dots, d_{364}\}$  into  $k$  clusters containing identical daily load time series, where each historical segment (day  $x$ ) is represented by a 24 h consumption vector  $d_x = \{l_0, l_1, \dots, l_{23}\}$  with  $l_h$  being the consumption at hour  $h$ . From a statistical point of view the problem in this case is considered as an unsupervised time series classification problem where the number  $k$  is unknown. To achieve this challenge, two questions should be answered: which clustering technique should be chosen and, if appropriate, how many clusters should be created. Nevertheless, There are more than a unique answer to both questions as it is dependent on sensitive factors. In the light of previous conducted work on the Algerian natural gas consumption dataset [39], where three different clustering methods were applied to non-supervisely classify the load profiles, namely:  $K$ -means, Mixture of Hierarchical Gaussian Process (MHGP) method [40], which combines Gaussian processes (GPs) approach to model the time series and Dirichlet processes (DPs) to perform clustering. The third is a Hierarchical Based Clustering with Noise (HDBSCAN) [41]. The approach is a non parametric method similar to MHGP. By looking at enhancing the overall forecast quality through, firstly, data segmentation investigating several techniques,  $K$ -means seems to be the most suitable one for the present dataset. In addition,  $K$ -means proved their efficiency when it comes to clustering large databases, including Algerian electrical data load in Ref. [36] a time series very much similar to the present one and will be used here.

### 3.1.2. $K$ -means clustering

A centroid-based clustering  $K$ -means is applied to find the appropriate grouping of daily consumption profiles, giving  $C = \{c_1,$

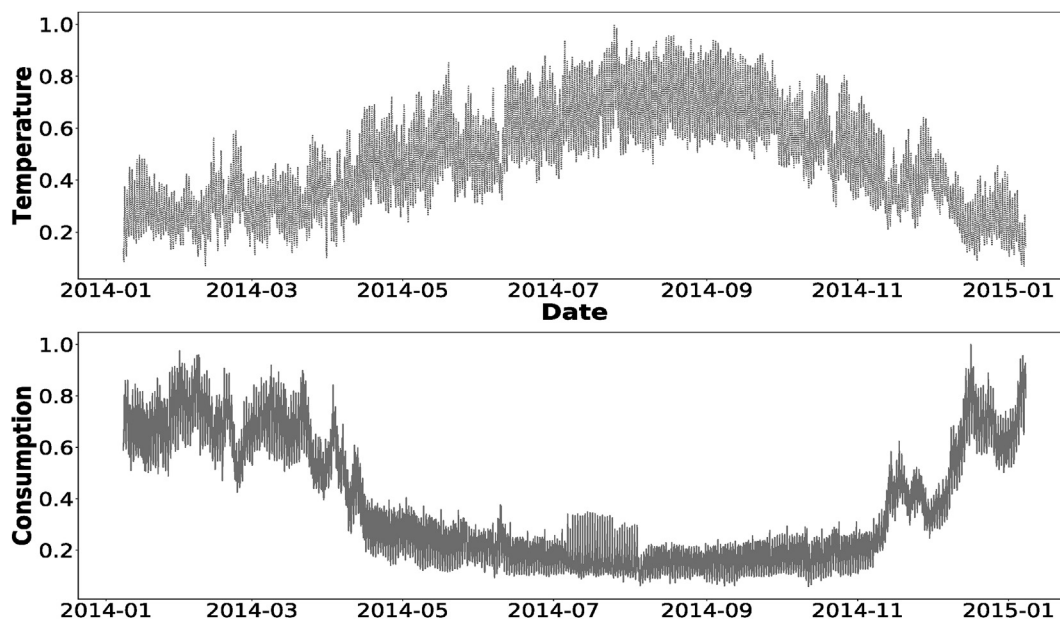


Fig. 2. Hourly recorder natural gas consumption and temperature during 2014.



$\dots, c_k\}$  denoting some partition of  $H_{2014}$  into  $k$  disjoint nonempty clusters, and  $n(c)$  denoting the number of data points assigned to a given cluster  $c$ . To determine the structure of day type, the  $k$ -means method follows an iterative algorithm of two steps [42] where every data observation  $d_i$  is assigned to its nearest cluster centroid  $\mu_i$  to form the clusters, then adjust the means  $(\mu_1, \dots, \mu_k)$  so that the total within-cluster sum of squared  $\varphi$  is minimized given by the distortion in Equation (1), where  $D$  is the Euclidean distance between each point to its centroid. This process keeps converging until the assignments are unchanged so the means remains unmoved when updated

$$\varphi(C) = \sum_{c=1}^K \sum_{i \in c} D(d_i, \mu_i) \quad (1)$$

Beginning with a random selection of  $k$  means, the data points assignment occurs according:

$$c_i = \underset{k}{\operatorname{argmin}} \{D^2(\mu_k - d_i)\} \quad (2)$$

where  $c_i$  is the index of the cluster which the point  $d_i$  is assigned to. The means are updated based on the new partition through:

$$\mu_c = \frac{1}{n(c)} \sum_{i \in c} d_i \quad (3)$$

### 3.2. Artificial neural network for natural gas forecasting

Recently, there has been a significant interest in the development of neural network algorithms for various applications. This has successfully brought ANNs on the top of numerous fields when it comes to solve non-linear problems. Because of the ability to approximate any measurable non-linear function to a certain desired degree of accuracy (Cybenko theorem [43]), ANNs are proven powerful universal approximators [44]. For function approximation, two major types of ANNs are used: feed-forward and recurrent architectures. Nowadays, Convolutional Neural Networks (CNN) are the most prominent model in the feed-forward variations, where they are destined and applied excessively for computer vision and pattern recognition tasks [45]. However, CNNs are highly profitable when the problem characteristics are not well

defined as they are extracted in the first phase of the training algorithm. This is not the case when it comes to a well defined function approximation like in natural gas load forecasting, where the regression input vector (characteristic vector) is known a priori. On the other hand, another type of RNN is LSTM, firstly introduced by Hochreiter and Schmidhuber where they demonstrated in Ref. [46] that LSTM can help solving many unsolved tasks using standard learning algorithms for RNN, especially for sequence processing tasks.

#### 3.2.1. MLP for daily consumption profile estimation

In this particular part of the study, a MLP is developed to estimate the next day consumption profile, playing the role of FM, by properly selecting a LSTM model to handle the day ahead natural gas consumption prediction. The FM-MLP model is characterised by a network of an input layer, two hidden layers and an output layer. The number of neurons in the input layer is equal to the number of the selected variables, where a different input sample sizes are considered to examine the effect of their selection on the estimation accuracy. The output layer consists of numerous neurons with respect to the recognized daily consumption profiles (number of clusters). The MLP topology is presented in the right side of Fig. 3, where the neurons output is forwarded to the successive layer, the output of the  $j^{\text{th}}$  unit is computed as follows:

$$y_j = f(z_j) \text{ where } z_j = \sum_{i=1}^I w_{ji} y_i + w_{j0} \quad (4)$$

where  $w_{ji}$  is the weighted connection between the  $i^{\text{th}}$  neuron to the  $j^{\text{th}}$  neuron in the next layer, and  $w_{j0}$  corresponds to the bias which is considered as an external neuron input. Here,  $I$  denotes the number of neurons in the source layer and  $f$  represents the activation logarithmic sigmoidal function used in this study:

$$\text{sigmoid} : f(x) = \frac{1}{1 + e^{-x}} \quad (5)$$

MLP training is performed using the BP algorithm, and is accomplished through adjusting the gradient weights to improve the ANN performance, in other term, minimizing the sum squared error which is calculated by:

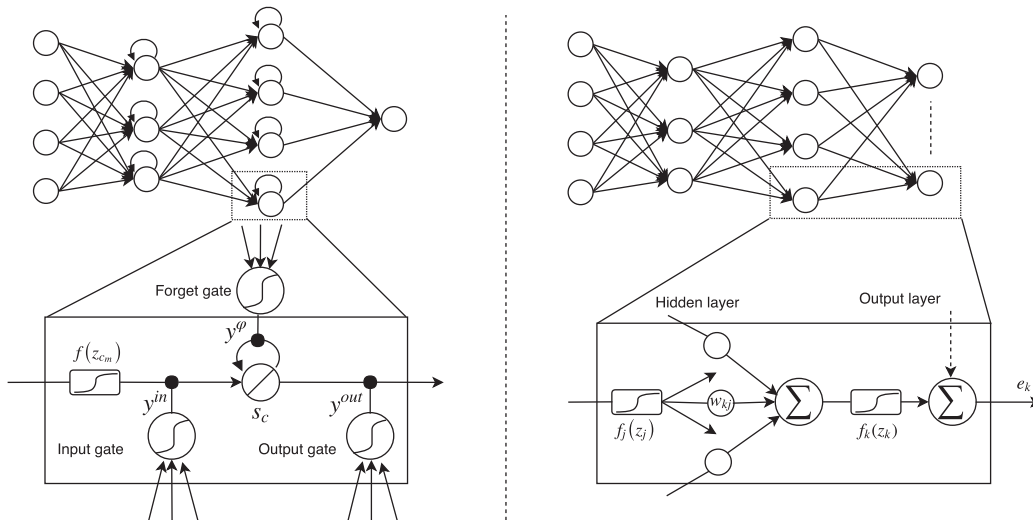


Fig. 3. An LSTM memory block with one cell (left side). An MLP network structure with information processing through neurons (right side).

$$E(t) = \frac{1}{2} \sum_{k=1}^K e_k^2 = \frac{1}{2} \sum_{k=1}^K (d_k - o_k)^2 \quad (6)$$

where  $E$  denotes the summed square error,  $k$  is the output layer neurons count,  $d_k$  and  $o_k$  represent the  $k^{th}$  target output and the network output respectively.

During the online training, the network weights are updated every time a data pattern is processed. The update process follows the idea of taking a step in the direction of the steepest descent, which is the direction of the negative gradient, to reach a minimum [47]. Thus, the weight update is given by:

$$w_{ji}(t+1) = w_{ji}(t) + \Delta w_{ji}(t) \quad (7)$$

and

$$\Delta w_{ji}(t) = -\mu \frac{\partial E}{\partial w_{ji}} = \mu \delta_j y_i \quad (8)$$

In summary, applying the chain rule using the learning rate  $\mu$  which is defined in the beginning of the training cycle, and thus, controls the training speed and stability of the network [48], the gradient of the cost function with respect to weights is then:

$$\text{for weights in the output layer : } \delta_k = 2f'_k(z_k)e_k \quad (9)$$

$$\text{for weights in the hidden layer : } \delta_j = 2f'_j(z_j) \sum_{k=1}^K \delta_k w_{kj} \quad (10)$$

### 3.2.2. LSTM for natural gas consumption forecasting

RNNs are inherently different compared to MLP. Even if the forward pass remains the same, with the exception that activations arise at the hidden layer from both the external input of the neuron itself and the previous time-step hidden layer activations. This way the network can establish the temporal correlations between previous information and the current state.

As stated before in section. 1, LSTMs have been found to outperform other traditional RNNs on tasks involving long time lags, their architecture permits to perform many more successful runs and much faster learning compared with Real Time Recurrent Learning (RTRL), Back Propagation Through Time, Recurrent Cascade-Correlation and Elman Nets [46]. Once daily consumption profiles are regrouped, several LSTM models are trained to learn the data for each cluster, the FM-MLP takes charge of selecting the right LSTM model in order to handle the forecasting task, which means predicting all hourly loads for the corresponding cluster. Typically, LSTM differs from the standard ANNs with its hidden layer units, where the summation units are replaced instead, by memory blocks. As shown in the left side of Fig. 3, that illustrates an LSTM memory block with a single cell, each block contains one or more self-connected memory cells and three sources of inputs: the input  $z_m^{in}$ , output  $z_m^{out}$  and the cell itself  $z_m^o$ . Each source is squashed with an activation function known as a gate that provides continuous regulators of write, read and reset operations for the cells. Consequently, the calculation of  $y_m$  in Equation (4) is replaced with the following sequence of equations

$$z_{cm}(t) = \sum_{m=0}^k w_{cmj} y_j(t-1) \quad (11)$$

where  $z_{cm}(t)$  is the network cell input which is firstly calculated

during each forward pass, then

$$z_m^{in}(t) = \sum w_{mj}^{in} y_j(t-1) ; y_m^{in}(t) = f_m^{in}(z_m^{in}(t)) \quad (12)$$

$$z_m^o(t) = \sum w_{mj}^o y_j(t-1) ; y_m^o(t) = f_m^o(z_m^o(t)) \quad (13)$$

$$z_m^{out}(t) = \sum w_{mj}^{out} y_j(t-1) ; y_m^{out}(t) = f_m^{out}(z_m^{out}(t)) \quad (14)$$

through the current study  $m$  refers to memory block with only one cell  $c_m$ , see Ref. [49] for details. Moreover,  $s_{cm}$  indexes the cell state of the  $m^{th}$  memory block which updated according to Equation (15)

$$s_{cm}(t) = y_m^o(t) s_{cm}(t-1) + y_m^{in}(t) f(z_{cm}(t)) \quad (15)$$

with

$$s_{cm}(0) = 0. \quad (16)$$

LSTM network training is a fusion of BP for output units and gate weights, and quietly modified and truncated version of RTRL for input weights, input gates and forget gates. Gradient truncates after one time-step and not by the flow of activation around the recurrent connection, thereby mitigating the gradient vanishing and exploding problem [50]. This eases the implementation of the algorithm, which is an important property for tasks such as time series prediction [51]. For a full derivation of the algorithm see Ref. [52].

For the output units, weight changes via gradient descent given by Equations (7)–(9), output gate weights are also obtained by standard back propagation

$$\Delta w_{mj}^{out}(t) = \mu \delta_m^{out}(t) y_j(t) \quad (17)$$

$$\delta_m^{out}(t) \stackrel{tr}{=} f_m^{out}(z_m^{out}(t)) \left( \sum s_{cm}(t) \sum w_{kc_m} \delta_k(t) \right) \quad (18)$$

here  $\stackrel{tr}{=}$  represents error truncation.

$$\frac{\partial s_{cm}(t)}{\partial w_{mj}^{in}} \stackrel{tr}{=} \frac{\partial s_{cm}(t-1)}{\partial w_{mj}^{in}} y_m^o(t) + f(z_{cm}(t)) f_m^{in}(z_m^{in}(t)) y_j(t-1) \quad (19)$$

$$\frac{\partial s_{cm}(t)}{\partial w_{cmj}} \stackrel{tr}{=} \frac{\partial s_{cm}(t-1)}{\partial w_{cmj}} y_m^o(t) + f'(z_{cm}(t)) y_m^{in}(t) y_j(t-1) \quad (20)$$

$$\frac{\partial s_{cm}(t)}{\partial w_{mj}^o} \stackrel{tr}{=} \frac{\partial s_{cm}(t-1)}{\partial w_{mj}^o} y_m^o(t) + s_{cm}(t-1) f_m^o(z_m^o(t)) y_j(t-1) \quad (21)$$

internal state error  $e_{s_{cm}}$  is calculated separately for each memory cell in order to calculate weights changes

$$e_{s_{cm}}(t) \stackrel{tr}{=} y_m^{out}(t) \left( \sum w_{kc_m} \delta_k(t) \right) \quad (22)$$

Weights corresponds to connections to the input, the cell and the forget gates are updated using the partials from Equations (19)–(21):

$$\Delta w_{cm}(t) = \mu e_{s_{cm}}(t) \frac{\partial s_{cm}(t)}{\partial w_{cmj}} \quad (23)$$

$$\Delta w_{mj}^{in}(t) = \mu e_{s_{cm}}(t) \frac{\partial s_{cm}(t)}{\partial w_{mj}^{in}} \quad (24)$$

$$\Delta w_{mj}^{\phi}(t) = \mu e_{s_{cm}}(t) \frac{\partial s_{cm}(t)}{\partial w_{mj}^{\phi}} \quad (25)$$

### 3.3. Experiments

#### 3.3.1. Clustering

In order to use the K-means clustering method it is necessary to initialize the number of clusters  $k$  in advance. Many criteria could be used to find the optimal count of  $k$  whether by considering compactness, separation or both in terms of summation or ratio. On one hand, as clustering evaluation indices based on within and between-cluster distances, the Silhouette index determines the optimal number of clusters by maximizing the value of the pairwise difference of between and within-cluster distances [53], the Calinski-Harabasz criterion validates the clustering by calculating the average between- and within-cluster sum of squares [54], the Davies-Bouldin index determines the best partition by choosing the minimum ratio of within-cluster scatter to between-cluster separation [55]. On the other hand, some criteria focus only on one aspect, such as the R-squared index that validates the clustering by measuring the homogeneity level between clusters [56]. Since the main purpose of the clustering phase in this study is regrouping the similar load profiles, it is more significant to validate the clusters by looking at the total within-cluster sum of squares without considering the between-cluster separation. After conducting several clustering processes with different count of groups, and according to the Elbow method [57], 3 clusters was selected as the best choice. In addition to these three identified classes ( $c_1, c_2$  and  $c_3$ ), the Algerian natural gas load behaves differently and the consumption patterns are sensibly different to nominal consumption in special periods such as the month of Ramadan and the special period of national and religious holidays labeled  $c_4$  and  $c_5$  respectively.

Once the clustering process is accomplished, 365 daily time series  $d_x[l_0...l_{23}]$  in the dataset  $H_{2014}$  are labeled with its correspondent cluster  $c_i$ . Fig. 4 shows 20 daily load samples from each cluster.

Table 2 shows how days included in each cluster spread across all four seasons. It can clearly be seen, that cluster  $c_1$  covers most of the winter observations with some days belonging to autumn. In addition, cluster  $c_2$  is composed of two weeks from the spring season and about a month of autumn days. Cluster  $c_3$  contains the rest of observation of the year with the exception of special days and periods. In order to complete the classification, cluster  $c_4$  is a subset of  $c_3$  representing 29 days of the month of Ramadan. For being a group of holidays only, cluster  $c_5$  lays on a very limited count of observation across the year.

#### 3.3.2. Neural networks for natural gas consumption modeling

After daily consumption profiles identification is performed, the natural gas consumption profiles data is then divided into three

**Table 2**  
Season's day count per cluster.

| Seasons   | Winter | Spring | Summer | Autumn |
|-----------|--------|--------|--------|--------|
| Cluster 1 | 82     | 0      | 0      | 18     |
| Cluster 2 | 2      | 15     | 0      | 27     |
| Cluster 3 | 0      | 74     | 64     | 40     |
| Cluster 4 | 0      | 0      | 29     | 0      |
| Cluster 5 | 4      | 4      | 1      | 5      |

These numbers represent days included in each cluster.

randomly drawn subsets: the first set, contains 60% of the input data is used for training to adjust the model parameters. The second set representing 20% of the input data, is used for the validation to ensure avoiding overfitting and the remaining 20% are saved and used for tests in order to assess the quality of the proposed approach. The evaluation using the first two sets was carried out by each model, representing each cluster, separately, while the third set is used to test the performance of the FM-MLP and LSTMs models combination.

In addition to historical and exogenous attributes, additional types of calendar information is included in the experiments. The first attribute is a day indicator identifying the day of the week for each day forecast. This could help distinguishing between working days and holidays and distinguishing between the first and last days of the week. Fig. 5 shows the variety of the daily average load per week on the generated clusters. The second calendar variable is an hourly indicator which captures the periodic pattern during the day. For representing long term changes, a monthly indicator was used.

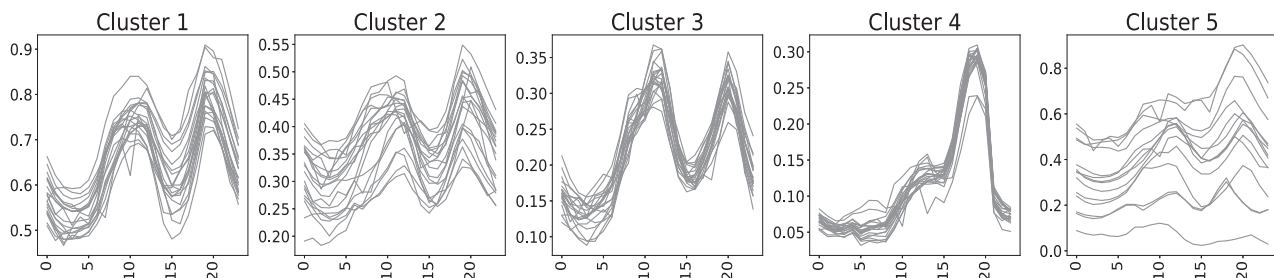
Dealing with two distinct tasks leads to the necessity of using different measures to evaluate the model's quality. The consumption profiles classification is formulated as a multi-type classification problem. The accuracy of the FM-MLP model is characterised by the number of the correct predictions  $cp$  divided by  $p$  the total number of predictions made, multiplied by 100 to represent it in a percentage:

$$Accuracy = \frac{cp}{p} 100 \quad (26)$$

As sigmoidal activation function is adopted for the output from the FM-MLP model which means that the model outcome is a probability between 0 and 1. In order to access these results precisely, a cross entropy loss or log loss is also used by comparing the outputs  $o_i$  with the targets  $d_i$  according:

$$\log loss = \frac{1}{l} \sum_{i=1}^l [d_i \log(o_i) + (1 - o_i) \log(1 - o_i)] \quad (27)$$

In order to check the accuracy level of the forecasting models, the Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) criteria



**Fig. 4.** Average load for the obtained clusters.

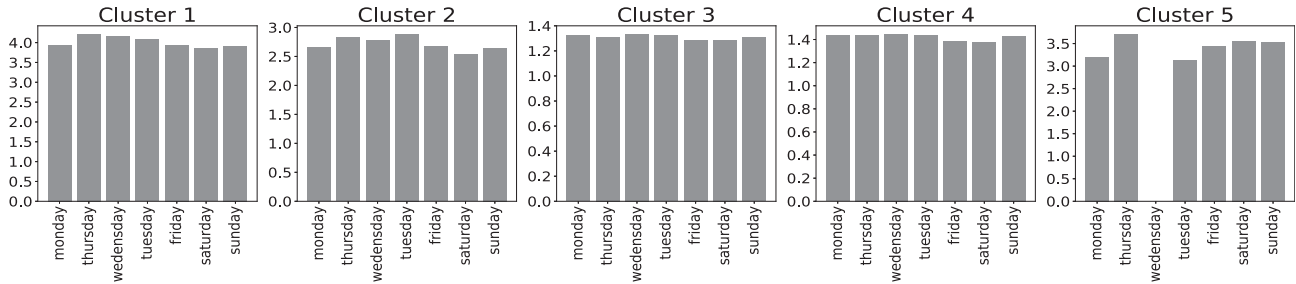


Fig. 5. The daily average natural gas consumption per week.

Table 3

Summary of used inputs, FM-MLP structure and learning parameters.

| Inputs                       |             |                             | Structure                    | Configuration |               |        |
|------------------------------|-------------|-----------------------------|------------------------------|---------------|---------------|--------|
| Lagged load                  | Temperature | Calendar variables          |                              | Activation    | Learning rate | Epochs |
| $[l_1, \dots, l_{24}]^{i-1}$ | $tempMax^i$ | day of week $[0, 1]_7$      | 20 nodes in 1st hidden layer | sigmoid       | 0.001         | 1000   |
| $[l_1, \dots, l_{24}]^{i-2}$ | $tempMin^i$ | month of year $[0, 1]_{12}$ | 10 nodes in 2nd hidden layer |               |               |        |

$$MAPE = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{a_i - p_i}{a_i} \right| \quad (28)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |a_i - p_i| \quad (29)$$

$$RMSE = \sqrt{\sum_{i=1}^N \frac{(a_i - p_i)^2}{N}} \quad (30)$$

are applied as a benchmark calculated according to the following formulas [58], where  $N$  denotes the number of predicted samples,  $a_i$  and  $p_i$  are the actual and predicted values respectively.

In order to achieve a better calculation of network weights, using an appropriate approach to normalize the natural gas consumption data is critical before feeding it to the LSTM networks. The outputs and inputs data are normalized to an interval between  $[0, 1]$ . Hence, a value of  $x$  is normalized to  $x'$  by computing:

$$x' = \frac{x}{x_{max}}, \quad (31)$$

where  $x'$  is the new value,  $x$  is the old value and  $x_{max}$  is the largest variable value in the year.

### 3.3.3. FM-MLP model

Due to the superiority of the four layers feed-forward network over the network containing only three layers for the training of input-target pairs highlighted in Ref. [59], a four layer topology is adopted in the current study. All experiments are performed with this four layers network structure in order to estimate the next day consumption profile. Because of the prior knowledge of last two added consumption profiles (included in clusters 4 and 5), there will be no need for their estimation. Therefore, the FM-MLP will have only 3 outputs representing the first 3 clusters.

The overall classification model structure and configuration is summarized in Table 3.

During the training stage of the proposed FM-MLP, multiple variables are taken into account: lagged consumption values

$[l_1, l_2, \dots, l_{24}]^{i-1}$  and  $[l_1, l_2, \dots, l_{24}]^{i-2}$  that represents the 48 hourly loads preceding the estimated day  $d_i$ , the maximum and minimum temperature of the forecast day forecast. Calendar characteristics are also considered during the classification procedure and used in term of one-hot-encoding  $[0, 1]$ .<sup>1</sup>

Table 4 shows the results obtained by the FM-MLP during training, validation and test phases. Based on the selected inputs, the FM-MLP shows a very powerful ability to estimate the next day consumption profile where only a single miss-classed estimation case has been recorded during the test phase.

### 3.3.4. LSTM models

To model the natural gas forecasting, several LSTM models representing the number of clusters were trained with the same variables used in the consumption profiles classification step. Enabling the model to capture the conditional dependencies between successive hourly consumptions over time, several experiments with different time lags was achieved as presented in Table 5. After evaluating various combinations of 1, 2, 4, 6, 8, 10 and 12 lagged load, it is noticeable that the best lag length windows varies from a cluster to another.

Beside the selected time lags and based on the autocorrelation analysis, two more previous hourly loads were added: ( $l_{t-24}$  and  $l_{t-168}$ ) representing the load at the same time of the previous day and the last week load at the same hour. This will help the model to keep real input values even for the case of multi step ahead forecasting. As weather information, actual estimated hourly temperatures, maximum and minimum temperature were used ( $temp_t$ ,  $tempMax_t$  and  $tempMin_t$  respectively). In addition to calendar information that has been used to train the FM-MLP, an hour indicator is also used, totalizing 60 possible input combination for LSTM models construction.

Another experiment was conducted to determine the most appropriate inputs. In order to enhance the prediction accuracy, these inputs are varied depending on each cluster's nature. However, the experiments are initialized using an input vector containing only the lagged load values. Then, as indicated in Table 6,

<sup>1</sup> Day of the week information consists in a 7 dimension vector with a values of 1 for the day  $d_i$  to forecast and zeros elsewhere, the month indicator is also used in the same manner with a 12 attributes vector.



**Table 4**

Accuracy, log loss and number of wrongly estimated profiles obtained on training, validation and test process.

| Results | Training |                      |                | Validation |                      |                | Test     |                      |                |
|---------|----------|----------------------|----------------|------------|----------------------|----------------|----------|----------------------|----------------|
|         | accuracy | log loss             | miss estimated | accuracy   | log loss             | miss estimated | accuracy | log loss             | miss estimated |
| FM-MLP  | 100%     | $1.59 \cdot 10^{-6}$ | 0              | 100%       | $4.58 \cdot 10^{-5}$ | 0              | 98.43%   | $5.90 \cdot 10^{-2}$ | 1              |

**Table 5**

Various combinations of 1, 2, 4, 6, 8, 10 and 12 time lags were evaluated using training sets.

| Models | 1 lag        | 2 lags | 4 lags | 6 lags | 8 lags | 10 lags      | 12 lags     |
|--------|--------------|--------|--------|--------|--------|--------------|-------------|
| LSTM 1 | 7.14         | 7.64   | 7.55   | 7.95   | 7.71   | 7.04         | <b>5.66</b> |
| LSTM 2 | <b>10.61</b> | 10.97  | 12.31  | 12.25  | 13.06  | 11.78        | 10.77       |
| LSTM 3 | 9.90         | 10.65  | 10.83  | 10.81  | 10.43  | 11.77        | <b>9.41</b> |
| LSTM 4 | 7.94         | 7.84   | 8.94   | 8.05   | 8.90   | 8.66         | <b>7.77</b> |
| LSTM 5 | 15.53        | 17.77  | 16.66  | 19.41  | 18.91  | <b>13.86</b> | 16.46       |

Bold underlined values represent the best performance for each model.

other variables were added sequentially to observe their impact on the model's prediction accuracy.

Despite the good performance achieved during the training process, generalization and good performances on test datasets are not straightforward. The network weights are optimized according to the training set, therefore, ANN models may experience overfitting during the learning process, causing poor generalization and thus degraded forecasting performances on test datasets [60]. This phenomena leads to choose the input combination used in (Exp 7) for LSTM 1 and LSTM 4 as well as inputs vector used in (Exp 6) for LSTM 2 and LSTM 5. As an example of overfitting cases, the MAPEs in test for LSTM 1 in (Exp 7) and (Exp 8) are 3.62% and 4.58% respectively, the input combination in (Exp 8) will thus be ignored.

The ANN topology represents the number of neurons per layer, the number of layers and how these neurons are connected. Generally, the neurons are arranged into one or two layers besides the input and the output layers [61]. Furthermore, finding the optimal number of hidden layers and neurons number within these layers must be decided precisely to construct a more accurate model, less sensitive to overfitting. However, there is no explicit method to determine these parameters. Hence, in addition to finding the most influential inputs for the forecasting, random search is used. Because a grid experiment with a fine-enough resolution for optimization would be prohibitively expensive, James Bergstra and Yoshua Bengio [62] recommended random search to find better models in most cases and in less computational time. By performing thirty experiments over the specified parameter values for each model, Table 7 shows different combinations of number of hidden neurons and values of learning rate to find the most successful model configuration.

**Table 6**

MAPE according to input combinations on training set.

| Experiments | Input variables |             |         |          |           | Models      |             |             |             |             |
|-------------|-----------------|-------------|---------|----------|-----------|-------------|-------------|-------------|-------------|-------------|
|             | Lagged load     | Temperature | Day Ind | Hour Ind | Month Ind | LSMT1       | LSMT2       | LSMT3       | LSMT4       | LSMT5       |
| Exp 1       | ✓               | ✓           |         |          |           | 6.02        | 7.64        | 9.10        | 7.46        | 10.62       |
| Exp 2       | ✓               |             | ✓       |          |           | 6.22        | 9.90        | 9.99        | 6.90        | 12.70       |
| Exp 3       | ✓               |             |         | ✓        |           | 4.60        | 5.76        | 8.25        | 6.23        | 61.42       |
| Exp 4       | ✓               |             |         |          | ✓         | 6.84        | 12.32       | 9.74        | 9.81        | 9.26        |
| Exp 5       | ✓               | ✓           | ✓       |          |           | 5.79        | 5.86        | 7.19        | 6.81        | 7.21        |
| Exp 6       | ✓               | ✓           |         | ✓        |           | 3.65        | <b>3.97</b> | 6.93        | 5.84        | <b>3.21</b> |
| Exp 7       | ✓               | ✓           | ✓       | ✓        |           | <b>3.52</b> | 2.65        | <b>5.78</b> | <b>4.40</b> | 2.54        |
| Exp 8       | ✓               | ✓           | ✓       | ✓        | ✓         | 2.13        | 5.03        | 7.65        | 4.24        | 60.66       |

Bold underlined values represent the best performance for each model.

**Table 7**

Parameters used in different experimental set-ups with 1000 epochs.

| Parameter              | Range              |
|------------------------|--------------------|
| 1st hidden layer nodes | (20–100)           |
| 2nd hidden layer nodes | (20–50)            |
| Learning rate          | ( $10^{-3}$ –0.01) |

## 4. Results and discussion

### 4.1. Models forecasting results

For the current forecasting problem, the performance across the most successful experiments related to the LSTMs architecture and the learning rate rule are presented in Table 8.

After selecting the most convenient inputs with the best model topology, determined through several epochs and ANN configurations, Table 9 reports the MAPE, MAE and RMSE for each LSTM model. Moreover, the model's performance on test set was evaluated based on the FM-MLP consumption profile estimations, thus, the MAPE, MAE and RMSE were obtained by the assigned LSTMs for all 5 clusters.

Fig. 6a and b clearly show a high similarity between actual and forecast load on the two clusters that cover the period of winter and the period of spring and autumn. In contrast, regardless the sort of inputs used to train the LSTM models and their structure, and due to the high non-linearity level in the natural gas consumption variation recorded in the summer, as presented in Fig. 6c, LSTM 3 forecast could not perfectly reflects the natural gas consumption. As shown in Fig. 6d and e, the corresponding LSTM models were not able to accurately estimate the daily peaks which consequently increase the MAPE, MAE and RMSE calculated on these two clusters.

**Table 8**

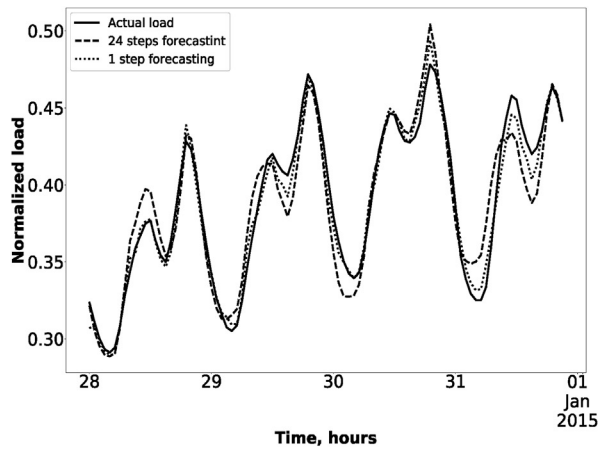
Parameters obtained by the random search optimization.

| Models | hidden layer size | Learning rate |
|--------|-------------------|---------------|
| LSTM 1 | (20–10)           | 0.001         |
| LSTM 2 | (20–10)           | 0.008         |
| LSTM 3 | (50–30)           | 0.002         |
| LSTM 4 | (90–30)           | 0.003         |
| LSTM 5 | (20–10)           | 0.001         |

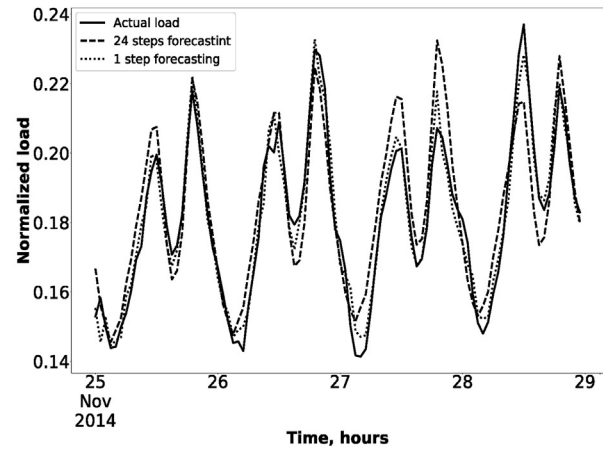
**Table 9**

MAPE and MAE of each cluster on the test sets.

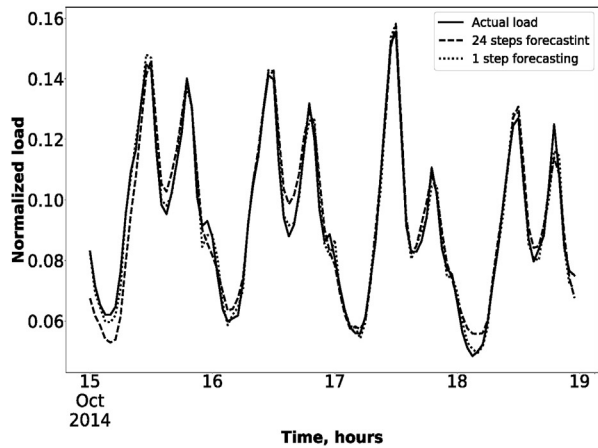
| Models    | MAPE         |      |                |      | MAE          |        |                |        | RMSE         |        |                |        |
|-----------|--------------|------|----------------|------|--------------|--------|----------------|--------|--------------|--------|----------------|--------|
|           | 1 step ahead |      | 24 steps ahead |      | 1 step ahead |        | 24 steps ahead |        | 1 step ahead |        | 24 steps ahead |        |
|           | Train        | Test | Train          | Test | Train        | Test   | Test           | Train  | Train        | Test   | Train          | Test   |
| Cluster 1 | 1.16         | 1.32 | 3.52           | 3.62 | 0.0040       | 0.0045 | 0.0119         | 0.0125 | 0.0060       | 0.0157 | 0.0060         | 0.0158 |
| Cluster 2 | 1.65         | 1.93 | 3.97           | 4.48 | 0.0034       | 0.0040 | 0.0086         | 0.0094 | 0.0048       | 0.0113 | 0.0054         | 0.0120 |
| Cluster 3 | 2.88         | 3.40 | 5.78           | 6.88 | 0.0024       | 0.0032 | 0.0051         | 0.0062 | 0.0033       | 0.0071 | 0.0043         | 0.0087 |
| Cluster 4 | 2.55         | 3.32 | 4.40           | 5.02 | 0.0018       | 0.0025 | 0.0035         | 0.0040 | 0.0036       | 0.0059 | 0.0025         | 0.0045 |
| Cluster 5 | 2.21         | 3.83 | 3.21           | 5.11 | 0.0028       | 0.0087 | 0.0051         | 0.0113 | 0.0041       | 0.0063 | 0.0102         | 0.0133 |



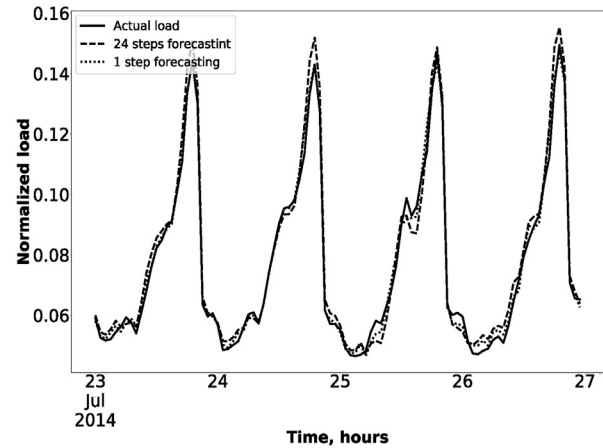
(a) Cluster 1



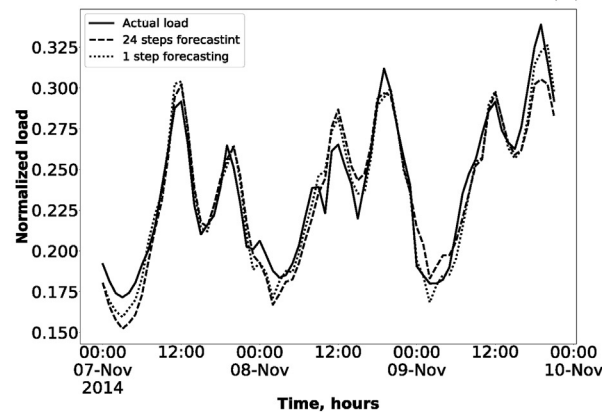
(b) Cluster 2



(c) Cluster 3



(d) Cluster 4



(e) Cluster 5

**Fig. 6.** 24 h forecast samples from each cluster test set.

## 4.2. Used benchmark models

To validate the enhancement of the proposed forecasting framework, the performance is compared with several strong state-of-the-art forecasting techniques, namely BP-ANN, LSTM-RNN, SARIMAX and MLR.

### 4.2.1. MLP-ANN and LSTM-RNN

The first considered benchmark models are global neural networks, with two considered types: a feed forward multi-layered ANN and the other is a LSTM RNN. These two proposed models have the same ANN structure containing 36 inputs (1, 24 and 168 lagged load, temperature and one-hot-encoding [0,1] variables to indicate the hour of the day and the day of the week). The ANNs structure is composed of two hidden layers with 20 neurons in the first and 10 in the second.

### 4.2.2. SARIMAX

Another important baseline approach is an extension of ARIMA models, which is a first class time series technique, introduced by Box and Jenkins [63]. The multivariate version of SARIMA has enhanced the ability to integrate explanatory (exogenous) variables and to add seasonal terms in order to increase the performance.

Seasonal ARIMA can be shown as SARIMA  $(p, d, q), (D, D, Q)_s$  where  $p, d$  and  $q$  referred to the order of autoregressive terms, the order of differencing and the number of moving averages terms in the non-seasonal and seasonal components of the model respectively. the term  $s$  represents the span of repeating the seasonal pattern which is equal to 24 in the current study referring to 24 hourly load per day.

Practically, most of time series are non-stationary and need to be transformed into a stationary ones before being used to regulate the model's parameters. Since the actual dataset as a load time series includes two types of seasonality, which is daily and weekly seasonality, two differencing were applied to obtain a stationary series, one at 24 lags and the second at 168 lags respectively. Also, a first order differencing was applied as well to overcome the trend in the load time series besides the differencing at seasonal lags. According to sample autocorrelation function (ACF) and partial autocorrelation function (PACF) for determining  $p$  and  $q$  variables, it is more evident to choose SARIMAX(1,1,1)(1,1,1)<sub>24</sub>, SARIMAX(2,1,1)(1,1,1)<sub>24</sub>, SARIMAX(3,1,1)(2,1,1)<sub>24</sub> and SARIMAX(6,0,1)(2,0,1)<sub>24</sub>, whereas the last

model presented the lowest MAPE among the candidate models.

### 4.2.3. Multiple linear regression

As for statistical techniques, MLR basis on estimating the unknown values of a variable from a set of other related known values. These models represent the gas consumption load as a linear function of several dependent and independent variables. However, despite the well pronounced uniform general trends of the data, non-linearities are still present and are difficult to capture. For instance, a MLR function provides and estimates load  $y$  at time  $t$  and  $n$  predictor variables  $x_i (i : 1, \dots, n)$  by:

$$y_t = \sum_{i=1}^n \theta_i x_i + \varepsilon \quad (32)$$

where  $\theta_i (i : 1, \dots, n)$  are regression parameters and  $\varepsilon$  is an error term.

## 4.3. Comparison and discussion

The benchmark models were trained using a set formed by combining the 5 training sets used for the LSTM models construction. Therefore, comparing their forecasting quality will be based on the 5 sets used to assess the LSTM models performance. Moreover, Each cluster deserves a particular consideration, and due to the high distinction in the recognized clusters over the year, weighted arithmetic mean is applied instead of the ordinary mean to calculate an average MAPE, MAE and RMSE that represents the overall error. Equation (33) expresses the weighted average of an error ( $\bar{E}$ ), where  $w_i$  is the number of days counted in each cluster  $c_i$ .

$$\bar{E} = \frac{1}{N} \sum_{i=1}^5 (w_i E_i) \quad (33)$$

According to the results presented in Table 10, the proposed approach had the lowest average MAPE of 24 steps ahead compared with the four benchmark methods. Additionally, the weighted average MAPE obtained by the proposed approach on test is 5.87%, which is an improvement over the global MLP and LSTM, SARIMAX and MLR by 16.42%, 14.41%, 13.50% and 56.75% respectively. Despite the high performance of global ANN approaches (MLP and LSTM), it is noticed that the proposed forecasting approach made a significant

**Table 10**

Performance of the proposed approach compared with MLP-ANN, LSTM-RNN, SARIMAX and MLR for one day ahead NG forecasting.

|      |           | Pro.app       |               | MLP    |        | LSTM   |        | SARIMAX |        | MLR    |        |
|------|-----------|---------------|---------------|--------|--------|--------|--------|---------|--------|--------|--------|
|      |           | Train         | Test          | Train  | Test   | Train  | Test   | Train   | Test   | Train  | Test   |
| MAPE | cluster 1 | 3.52          | 3.62          | 3.97   | 5.36   | 3.25   | 5.49   | 4.37    | 3.39   | 4.69   | 4.48   |
|      | cluster 2 | 3.97          | 4.48          | 4.69   | 5.26   | 3.91   | 5.34   | 6.11    | 5.03   | 7.55   | 5.67   |
|      | cluster 3 | 5.78          | 6.88          | 6.62   | 6.84   | 6.30   | 6.38   | 9.27    | 7.12   | 9.94   | 9.12   |
|      | cluster 4 | 4.40          | 5.02          | 8.79   | 7.77   | 8.11   | 7.48   | 9.89    | 11.13  | 19.20  | 23.30  |
|      | cluster 5 | 3.21          | 5.11          | 7.71   | 8.55   | 6.98   | 10.88  | 10.47   | 8.68   | 8.64   | 10.06  |
|      | average   | <b>4.73</b>   | <b>5.48</b>   | 5.87   | 6.38   | 5.34   | 6.27   | 7.64    | 6.22   | 8.89   | 8.59   |
| MAE  | cluster 1 | 0.0119        | 0.0125        | 0.0133 | 0.0187 | 0.0112 | 0.0192 | 0.0152  | 0.0115 | 0.0160 | 0.0153 |
|      | cluster 2 | 0.0086        | 0.0094        | 0.0103 | 0.0110 | 0.0088 | 0.0110 | 0.0144  | 0.0104 | 0.0176 | 0.0121 |
|      | cluster 3 | 0.0051        | 0.0062        | 0.0059 | 0.0061 | 0.0054 | 0.0059 | 0.0076  | 0.0064 | 0.0082 | 0.0088 |
|      | cluster 4 | 0.0035        | 0.0040        | 0.0075 | 0.0057 | 0.0068 | 0.0056 | 0.0079  | 0.0074 | 0.0155 | 0.0155 |
|      | cluster 5 | 0.0051        | 0.0113        | 0.0138 | 0.0194 | 0.0119 | 0.0231 | 0.0187  | 0.0205 | 0.0164 | 0.0218 |
|      | average   | <b>0.0072</b> | <b>0.0083</b> | 0.0078 | 0.0106 | 0.0087 | 0.0107 | 0.0109  | 0.0088 | 0.0123 | 0.0119 |
| RMSE | cluster 1 | 0.0157        | 0.0158        | 0.0175 | 0.0223 | 0.0149 | 0.0227 | 0.0202  | 0.0156 | 0.0212 | 0.0193 |
|      | cluster 2 | 0.0113        | 0.0120        | 0.0141 | 0.0142 | 0.0118 | 0.0137 | 0.0191  | 0.0128 | 0.0224 | 0.0151 |
|      | cluster 3 | 0.0071        | 0.0087        | 0.0081 | 0.0080 | 0.0075 | 0.0079 | 0.0111  | 0.0085 | 0.0110 | 0.0116 |
|      | cluster 4 | 0.0059        | 0.0045        | 0.0109 | 0.0071 | 0.0104 | 0.0071 | 0.0127  | 0.0090 | 0.0208 | 0.0189 |
|      | cluster 5 | 0.0063        | 0.0133        | 0.0194 | 0.0223 | 0.0164 | 0.0257 | 0.0272  | 0.0259 | 0.0218 | 0.0258 |
|      | average   | <b>0.0092</b> | <b>0.0108</b> | 0.0140 | 0.0147 | 0.0122 | 0.0154 | 0.0180  | 0.0143 | 0.0194 | 0.0181 |

Bold underlined values represent the best average performance.

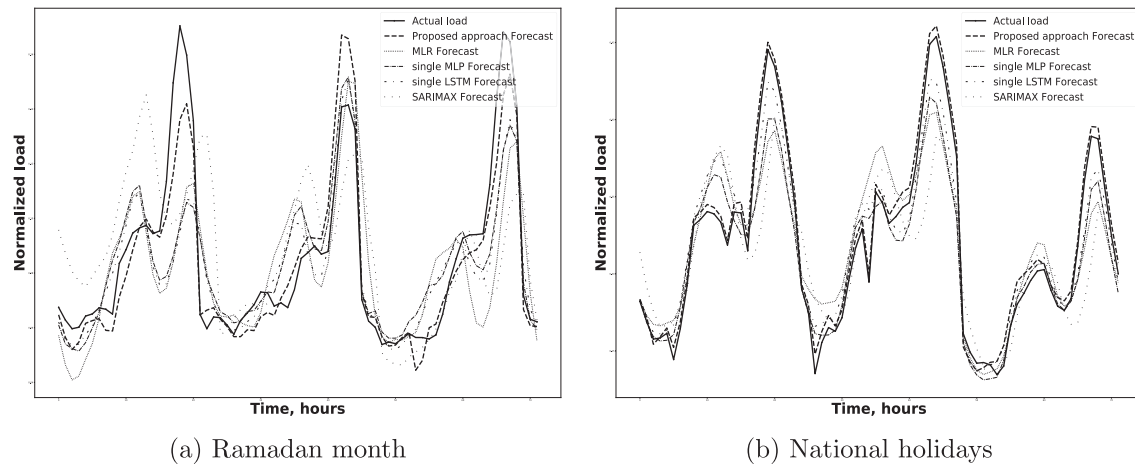


Fig. 7. Comparison between actual and predicted natural gas consumption with the use of the proposed and benchmark approaches.

improvement on the forecast quality especially over the test dataset.

Considering only the forecasting performance, the proposed method has the minimum error including for special days when consumption curves change, this is evident in the case of national holidays or when the customers behavior changes due to occasional reasons. Moreover, all benchmark approaches show a weakness on these specific periods due to the high dependency of the previous loads. However, multiple LSTMs models are relatively accurate compared to a global model approach. Fig. 7 presents the advantage of the proposed approach upon other approaches on the national holidays and the beginning of the month of Ramadan.

## 5. Conclusion

The main contribution of this work consists in the developed two-stage Forecasting Monitoring Multi Layered Perceptron (FM-MLP) approach. After partitioning the Algerian natural gas hourly consumption data by applying *K*-means according to the consumption profiles, the FM-MLP is then trained based on the dataset segmentation for the purpose of estimating the next day consumption profile. The FM-MLP decides which LSTM recurrent model will handle best its forecast. The approach achieves 98.34% accuracy of classification of the natural gas consumption profile on the real test dataset.

Based on the obtained groups that contain similar consumption profiles and by using calendar and temperature information, multiple LSTM recurrent networks are constructed according to each cluster data. The performance of the developed approach is carefully validated and evaluated based on the gas consumption profile forecast made by the FM-MLP. The weighted average of the MAPE, MAE and RMSE obtained by the assigned LSTM models on the testing set are 5.48%, 0.0083 and 0.0108, respectively.

In order to validate the obtained results, the MLP ANN, LSTM RNN, SARIMAX and MLR models are employed. Comparing the predicted results with each benchmark method in terms of MAPE, MAE and RMSE, the developed approach shows improved accuracy. The superiority is especially pronounce in the periods with irregular gas consumption, such as during holidays and special days periods where daily natural gas consumption possesses a quite unique curve that could not be captured using a single general approach.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2019.04.075>.

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